MIL-C-3098/85A (USAF)
11 April 1975
SUPERSEDING
MIL-C-3098/85 (USAF)
23 August 1967

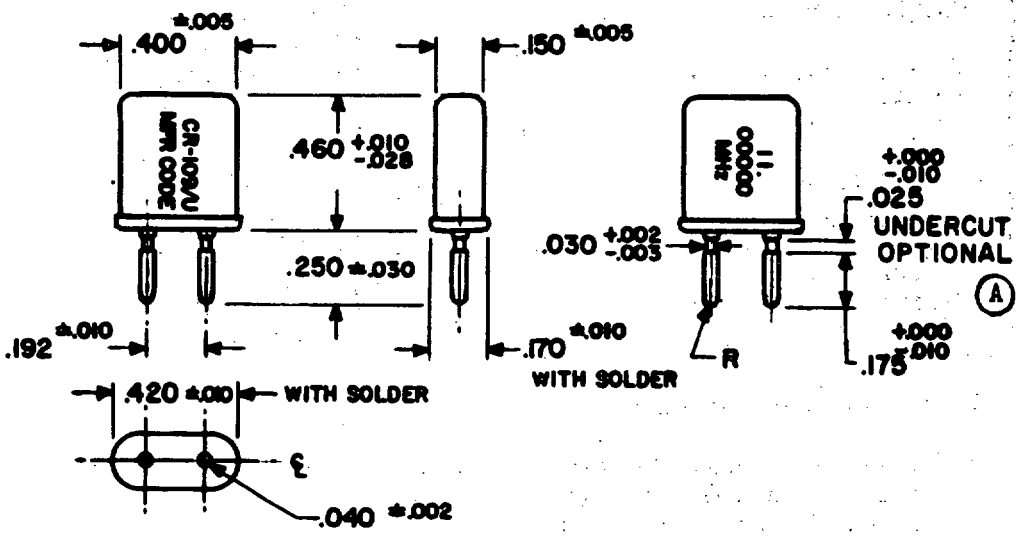
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-109/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Military Specification MIL-C-3098.

Pertinent characteristics: 10 to 11.5 MHz; fundamental; controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.150	3.81
.003	.08	.170	4.32
.005	.13	.175	4.45
.010	.25	.192	4.88
.025	.64	.250	6.35
.028	.71	.400	10.16
.030	.76	.420	10.67
.040	1.02	.460	11.68

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm TO THE NEAREST .01 mm.
3. THE TYPE-DESIGNATION, FREQUENCY AND CODE DESIGNATING CHARACTERS SHALL NOT BE LESS THAN .058 (.97 mm) IN HEIGHT AND NOT LESS THAN .027 (.69 mm) IN WIDTH.

(A) DENOTES CHANGE

FSC 5955

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TABLE I. Requirements

Requirement	Specified Value or Condition
Holder	HC-25/U, modified (MIL-H-10056/11)
(A) Dimensions, marking, and configuration	See Figure
Frequency range	10 to 11.5 MHz, inclusive
Frequency tolerances:	
Operating temperature range	±0.002 percent
Reference temperature	±0.002 percent
Room temperature	±0.008 percent
Frequency stability	±0.0005 percent
Mode of oscillation	Fundamental
Reference temperature	85° ±2°C
Temperature ranges:	
Operating (controlled)	80° to 90°, inclusive
Operable	-55° to +80°C, inclusive
Government reference standard test set	TS-330/TSM
Capacitance, load	32.0 ±0.5 pf
Rated drive level	2.5 ±0.5 mw
Calibration values	Resistance..... 11 ohms Crystal current.. 15 ma
Capacitance, shunt	7 pf, maximum
Equivalent resistance	22 ohms, maximum
Shock: 1/	
Frequency change permitted	±0.0004 percent
Equivalent resistance change permitted	±10 percent
Vibration: 2/	
Frequency change permitted	±0.0004 percent
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	0.0005 percent

1/ MIL-C-3098D, Method B.

2/ Method 201 of MIL-STD-202, double amplitude 0.036 inches or 10 g's whichever is the limiting value. Frequency shall be cycled between 10 and 500 cps in 15 minute cycles.

3/ Requirements Not Applicable: Reduced drive level; Heat cycling; Bond strength.

4/ These crystal units are for use in the Interceptor System, Aircraft and Weapon Control Type MA-1 and AN/ASQ-25. They were previously fabricated using Air Force Technical Order 12-1-20 and/or Electronic Communications, Inc. Drawing Number RD24300-1.

Custodian:
Air Force - 84

Review activities:
Air Force - 11, 85

User activities:
Air Force - 17, 19

Preparing activity:
Air Force - 84

Project No. (5955-F171-8)

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OMB Approval
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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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